

Table of Integrals

Powers of x .

$$\left. \begin{aligned} \int x^n dx &= \frac{1}{n+1} x^{n+1} + C \quad (n \neq -1) \\ \int \sqrt{x} dx &= \frac{2}{3} x^{3/2} + C \end{aligned} \right\} = \ln|x| + C$$

Basic Trigonometric Functions.

$$\left. \begin{aligned} \int \sin x dx &= -\cos x + C \\ \int \cos x dx &= \sin x + C \end{aligned} \right\} \begin{aligned} \int \sec^2 x dx &= \tan x + C \\ \int \sec x \tan x dx &= \sec x + C \end{aligned}$$

Advanced Trigonometric Functions.

$$\begin{aligned} \int \tan x dx &= \ln|\sec x| + C \\ \int \sec x dx &= \ln|\sec x + \tan x| + C \end{aligned}$$

Exponentials

$$\int e^x dx = e^x + C \qquad \int b^x dx = \frac{b^x}{\ln b} + C$$

Logs

$$\int \ln x dx = x \ln x - x + C$$

Quadratic Denominators (Inverse Trig. Functions)

$$\begin{aligned} \int \frac{1}{x^2 + a^2} dx &= \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C & \int \frac{1}{\sqrt{a^2 - x^2}} dx &= \arcsin\left(\frac{x}{a}\right) + C \\ \int \frac{1}{|x| \sqrt{x^2 - a^2}} dx &= \frac{1}{a} \operatorname{arcsec}\left(\frac{x}{a}\right) + C \end{aligned}$$

Substitution with Indefinite Integrals.

1. Compute the following integrals involving $\ln$

(A) $\int \frac{1}{x} dx$

(B) $\int \frac{x^3}{3x^4 + 5} dx$

(C) $\int \frac{(2 \ln x + 3)^4}{x} dx$

2. Compute the following integrals involving e^x

(A) $\int e^x dx$

(B) $\int x^3 e^{3x^4+5} dx$

(C) $\int \frac{e^{3x}}{2e^{3x} + 4} dx$

3. Compute the following integrals involving $\arctan$.

(A) $\int \frac{3}{x^2 + 1} dx$

(B) $\int \frac{2x}{x^2 + 1} dx$

(C) $\int \frac{2x + 3}{x^2 + 1} dx$

(D) $\int \frac{5}{4x^2 + 9} dx$